CURRICULUM VITAE

Prof. Muhittin Gökmen

MEF University
Faculty of Engineering
Department of Computer Engineering
Maslak 34396 İSTANBUL
Tel: +90 532 384 5161
E-mail: gokmenm@mef.edu.tr
Web: https://www.muhittingokmen.com/

Date and place of Birth :

August 1, 1959
Boyabat,Sinop TURKEY

Education:

- Ph.D.: University of Pittsburgh, Pittsburgh PA, USA, December 1990.
- MSc: University of Pittsburgh, Pittsburgh PA, USA, April 1986.
- MSc: Istanbul Technical University, Faculty of Electrical and Electronics, Department of Electronics and Communication, June 1984
- BSc: Istanbul Technical University, Faculty of Electrical and Electronics, Department of Electronics and Communication, June 1982

Areas of Interest:

- Computer Vision
- Image Processing
- Artificial Intelligence
- Pattern Recognition
- Digital Signal Processing

Professional Experience:

- 1986-1988 **Research Assistant**, Department of Electrical Engineering, University of Pittsburgh, USA
- 1988-1990 **Teaching Assistant**, Department of Electrical Engineering, University of Pittsburgh, USA
- 1990-1991 **Post-Doctoral Research Fellow**, Department of Electrical Engineering, University of Pittsburgh, USA

- 1992-1993 **Assistant Professor**, Department of Control&Computer Engineering, Istanbul Technical University
- 1993-1999 **Associate Professor**, Department of Computer Engineering, Istanbul Technical University
- 1999 -2015 **Professor**, Department of Computer Engineering, Istanbul Technical University
- 1993-1997 **Visiting Researcher,** Michigan State University, Department of Computer Science (one or two months per year).
- 1996 - 2010 **Founding Director** of ITU Multimedia Center
- 1996 - 2010 **Founding Director** of ITU Software Development Center
- 1997 - **Founding Director** of ITU-TSK Modelling and Simulation Lab.
- 2004 - 2018 **Founding Director of Divit Inc**., ITU Arı Technopark
- 2001- 2004 **Dean**, Faculty of Electrical & Electronics Engineering, Istanbul Technical University
- 2009- 2011 **Chair**, Department of Computer Engineering, Istanbul Technical University
- 2010- 2014, **İTÜ Senate member**, Representative of Computer and Informatics Faculty
- 2014- 2023, **Department Chair**, MEF University. Computer Engineering Department
- 2015-2023, **Director of Graduate School of Science and Engineering**. MEF University
- 2023-2024, **Sabbatical Leave**, University of Pennsylvania, Children Hospital of Philadelphia (CHOP), Center for Autism Research (CAR)
- 2025- To date, **Vice Rector for Research**, MEF University

Journal Papers:

1. C. C. Li, M. Gökmen, A. D. Hirschman and Y. Wang, "Information Preserving Image Compression for Archiving NMR Images," Computerized Medical Imaging and Graphics, vol.15, pp. 277-283, 1991.

2. M. Gökmen and R. W. Hall, "Parallel Shrinking Algorithms Using 2-Subfields Approaches," Computer Vision, Graphics and Image Processing, vol. 52, 1990, pp. 191-209.

3. M. Gökmen and C. C. Li, "Edge Detection and Surface Reconstruction Using Refined Regularization," IEEE Trans. Pattern Analysis and Machine Intelligence, vol.15, No.5, 1993, pp.492-499.

4. M.Gökmen, A.K. Jain, "Lambda Tau-Space Representation of Images and Generalized Edge Detection," IEEE Trans.on Pattern Analysis and Machine Intelligence, vol.19, No. 6, June 1997, pp.545-563.

5. A.Gültekin and M. Gökmen," Adaptive Shape from Shading," Turkish Journal of ELEKTRİK vol. 6, No. 2, 1998, pp. 61-73.

6. A.Yılmaz and M.Gökmen, "Eigenhills vs Eigenface and Eigenedge," Pattern Recognition, vol. 34, 2001, pp. 181-184.

7. A. Capar, B.Kurt, M.Gokmen, "Affine-Invariant Gradient Based Shape Descriptor", Machine Vision & Applications, vol.20,No:6,365-378, 2009,

8. B. Tunç, M.Gökmen,"Manifold Learning for Face Recognition Under Changing Illumination," Telecommunication Systems, Special Issue on Face Recognition, Vol.47, Iss.3-4, pp.185, 2011, ISSN: 10184864

9. B.Tunc, V.Dağlı, M.Gökmen, "Class Dependent Factor Analysis and Its Application to Face Recognition", Pattern Recognition, vol 45, no:12, 2012, pp.4092-4102.

10. A. Akyol, M.Gokmen," Super-Resolution Reconstruction of Faces by Enhanced Global Models of Shape and Texture," Pattern Recognition, 2012, vol 45, no:12, 2012, pp.4103-4116.

11. F.Kahraman, B. Kurt, M. Gökmen, "License Plate Character Segmentation Based on the Gabor Transform and Vector Quantization" Lecture Notes on Computer Science LNCS, Springer-Verlag, November 2003. pp 382-388.

12. A. Çapar, K. Taşdemir, Ö.Kılıç, M. Gökmen "A Turkish Handprint Character Recognition System" Lecture Notes on Computer Science (LNCS) , Springer-Verlag, November 2003, pp. 447-456.

13. I.K. Yalcin, M. Gökmen," License plate tracking from monocular camera view by condensation algorithm," Lecture Notes on Computer Science (LNCS-3645), Springer-Verlag, 2005. pp.860-869.

14. F. Kahraman, M. Gökmen, Illumination Invariant Face Alignment Using Multi-band Active Appearance Model., Lecture Notes on Computer Science (LNCS 3776) , Springer-Verlag, December 2005, pp. 118-127.

15. A. Çapar, B. Kurt, M. Gökmen, "Affine-Invariant Gradient Based Shape Descriptor," Lecture Notes in Computer Science, International Workshop on Multimedia Content Representation, Classification and Security, 2006, pp. 514-521.

16. M. Uzun Per, M. Gokmen,"Face Recognition with patch-based local Walsh transform," Signal Processing: Image Communication 61, 2018, pp. 85-96.

17. E. Başaran, M. Gökmen, ME Kamaşak, "An efficient multiscale scheme using local Zernike moments for face recognition," Applied Sciences, 2018, 8 (5), 827.

18. An efficient framework for visible-infrared cross modality person re-idenfication, Signal Processing: Image Communication, 2020, 87,115933

International Conference Papers:

1. C.C. Li, M. Gökmen, A. D. Hirschman And Y. Wang, "Information Preserving Image Compression For Archiving NMR Images," Proc. Of 9th International Conf. On Pattern Recognition, Rome, Italy, 1989, Pp. 1295-1299

2. M.Gökmen And C. C. Li, "Edge Detection With Iteratively Refined Regularization," Proc. Of 10th International Conf. On Pattern Recognition, Atlantic City, U.S.A., June 1990, Pp. 690-693.
3. M.Gökmen, "A Comparison Of Edge Detection Algorithms Based On Gaussian Filtering And Iteratively Refined Regularization," Proc. Of Bilkent International Conf. On New Trends İn Communication, Control And Signal Processing, Ankara, Turkey, July 1990, Pp. 1370-1377.
4. M.Gökmen And C. C. Li, "Edge Detection Using Refined Regularization," Proc. Of IEEE Computer Vision And Pattern Recognition Conference, Maui, Hawaii, June 3-6, 1991, Pp. 215-221.
5. M.Gökmen And C. C. Li, "Multiscale Edge Detection Using First Order R-Filter" Proc. Of Int. Conf. On Pattern Recognition ICPR92, The Hague, The Netherlands, August 1992, Pp. 307-310.
6. T.Acar Ve M. Gökmen, "Image Coding Using Weak Membrane Model Of İmages," Proc. Of Visual Communications And Image Processing'94, 25-29 September 1994, Chicago, USA, Pp. 1221-1230.
7. M.Gökmen And T.Acar," Multiscale Edge Integration By Weighted Accumulation, Proc. Of The Tenth International Symposium On Computer And Information Sciences (ISCIS X), Vol.1, Oct.30-Nov.1, 1995, Kusadasi Turkey, Pp. 111-118.
8. N.Ratha, T. Acar, M. Gökmen, A.K. Jain, "A Distributed Edge Detection And Surface Reconstruction Algorithm, "Proc. Computer Architecture For Machine Perception, CAMP'95, Italy, 1995, Pp. 149-154
9. M.Gökmen, A.K. Jain, "λ-τ Space Representation Of Images And Generalized Edge Detection," Proc. IEEE Conf. On Computer Vision And Pattern Recognition CVPR96, San Francisco, USA, 1996, Pp. 764-769.
10. M. Gökmen, İ. Ersoy, A. K. Jain, "Compression Of Fingerprint İmages Using Hybrid İmage Model", Proc. IEEE Intl. Conf. On Image Proc. ICIP'96, Vol.III, Pp.395-398, 1996.
11. A.Gültekin And M. Gökmen," Adaptive Shape From Shading," Proc. Of The Eleventh International Symposium On Computer And Information Sciences (ISCIS XI), Antalya, Turkey, November 6-8, 1996, Pp83-92.
12. B.Kurt, M.Gökmen And A.K. Jain, "Image Compression Based On Centipede Model," Proc. Of Int. Conf. On Image Analysis And Processing (ICIAP'97), Florence Italy, September 1997, Vol.I, Pp.303-310.
13. B.Kurt, M.Gökmen,"Two Dimensional Generalized Edge Detector," To Appear İn Proc. Of The 10th International Conference On Image Analysis And Processing, (ICIAP'99) , Venice Italy, September 1999.
14. A.Yılmaz And M. Gökmen, "Eigenhill Vs. Eigenface And Eigenedge," Proc. Of Int. Conf. On Pattern Recognition, ICPR'2000, Barcelone Spain, Sept. 2000.
15. F. Kahraman, M. Gökmen, "Face Recognition With The Use Of Principal Component Analysis Based On Artificial Neural Networks." 25th European Conference On Visual Perception, Glasgow, 2002.
16. E. Çayırcı, M. Gökmen, S. Oktuğ, T. Ayar, G. Cebiroğlu, Eryiğit, T. Çöplü, E.Okçu, İ. Akgün, "A Graphical Interface And A Data Filtering Scheme For Joint Theather Level Simulation", MSG-022 C3I And M&S Interoperability Conference, NATO Modelling And Simulation Co-Ordination Office (NMSCO), Antalya, 2003.
17. A.Çapar, K.Taşdemir, Ö.Kılıç, M.Gökmen "A Turkish Handprint Character Recognition System" 18th International Symposium On Computer And Information Sciences (ISCIS'03), Antalya TURKEY, 2003
18. S. Şener, H. Demirel, M. Gökmen, An Adaptive Modulation Technique For Spread Spectrum Video Watermarking, IASTED-SIP2004 ,
19. Sirin T., Saglam M.I., Erer I., Gokmen M., Edge Detection İn Images Using Clustering Algorithms, WSEAS Teleınfo 2005, Prague, Czech Republic, March 13-15, 2005.

20. Sirin T., Saglam M.I., Erer I., Gokmen M., Ersoy O., Edge Detection İn Multispectral Remote Sensing Image, RAST 2005, İstanbul, Turkey, June 09-11, 2005.

21. Sirin T., Saglam M.I., Erer I., Gokmen M., Ersoy O., A Comparative Evaluation Of Competitive Learning Algorithms For Edge Detection Enhancement?, EUSIPCO2005, Antalya,Turkey, September 4-8, 2005

22. Çapar, M. Gökmen, "Concurrent Segmentation And Recognition With Shape-Driven Fast Marching Methods", International Conference On Pattern Recognition ICPR'06, Hong Kong, 2006.

23. İ.K Yalçın ., M. Gökmen, "Integrating Differential Evolution And Condensation Algorithms For License Plate Tracking", International Conference On Pattern Recognition ICPR'06, Hong Kong, 2006.

24. F. Kahraman, M. Gökmen, "Illumination Invariant Three-Stage Approach For Face Alignment", Accepted To 13th International Conference On Image Processing (ICIP 2006), 08-11 October, 2006, Atlanta, USA.

25. Emre Bertan, Muhittin Gökmen, Guillaume Moreau, Automatic 3D Roof Reconstruction Using Digital Cadastral Map, Architectural Knowledge And An Aerial Image, IEEE Interntional Geosicence & Remote Sensing Symposium IGARSS'06, Denver Colorada, USA, 31-July-04 August 2006,

26. Mahmut Meral, Muhittin Gökmen, Isolated Word Recognition On Embedded System, WSEAS, Hangzhou, China, April 16-18, 2006.

27. F. Kahraman, M. Gokmen, S. Darkner, R. Larsen, An Active Illumination And Appearance (AIA) Model For Face Alignment, IEEE Conf. On Computer Vision And Pattern Recognition (CVPR 2007), IEEE Computer Society Workshop On Biometrics, Minneapolis, Minnesota, USA, 2007. [Pdf]

28. F. Kahraman, B. Kurt, M. Gokmen, Robust Face Alignment For Illumination And Pose Invariant Face Recognition, IEEE Conf. On Computer Vision And Pattern Recognition (CVPR 2007), IEEE Computer Society Workshop On Biometrics, Minneapolis, Minnesota, USA, 2007.

29. Kurt, Binnur; Gokmen, Muhittin; Goal Oriented Edge Detection Computer And Information Sciences, 2008. ISCIS '08. 23rd International Symposium On 27-29 Oct. 2008 Page(S):1 – 5

30. Kirtac, Kadir; Dolu, Onur; Gokmen, Muhittin; F**ace Recognition By Combining Gabor Wavelets And Nearest Neighbor Discriminant Analysis,** Computer And Information Sciences, 2008. ISCIS '08. 23rd International Symposium On 27-29 Oct. 2008 Page(S):1 – 5

31. Kahraman, Fatih; Mendi, C. Deniz; Gokmen, Muhittin;, **Image Frame Fusion Using 3D Anisotropic Diffusion,** Computer And Information Sciences, 2008. ISCIS '08. 23rd International Symposium On, 27-29 Oct. 2008 Page(S):1 – 6

32. Bayhan, Mehmet Serif; Gokmen, Muhittin; **Scale And Pose Invariant Real-Time Face Detection And Tracking,** Computer And Information Sciences, 2008. ISCIS '08. 23rd International Symposium On 27-29 Oct. 2008 Page(S):1 – 6

33. Akyol, Aydin; Gokmen, Muhittin; **Subspace Representation Of Registration And Reconstruction In Multi-Frame Super-Resolution,** Computer And Information Sciences, 2008. ISCIS '08. 23rd International Symposium On 27-29 Oct. 2008 Page(S):1 – 6

34. Kus, Merve Can; Gokmen, Muhittin; Etaner-Uyar, Sima;, **Traffic Sign Recognition Using Scale Invariant Feature Transform And Color Classification,** Computer And Information Sciences, 2008. ISCIS '08. 23rd International Symposium On 27-29 Oct. 2008 Page(S):1 – 6

35. Baris D. Ekinci, M. Gökmen, A Ball Tracking System For Offline Tennis Videos, Proc. Of The 1st WSEAS Int. Conf. On VISUALIZATION, IMAGING And SIMULATION (VIS'08), Bucharest, Romania, November 7-9, 2008,

36. G. Jun, J. K. Aggarwal, And M. Gokmen. Tracking And Segmentation Of Highway Vehicles In Cluttered And Crowded Scenes. IEEE Workshops On Applications Of Comp. Vision (WACV), 2008.

37. O. Dolu, K Kirtac, M Gokmen, Ensembled gabor nearest neighbor classifier for face recognition, . ISCIS 2009. 24th International Computer and Information Sciences, 2009

38. B.Tunc,S.K. Zhou, J. Park, M. Gökmen, Context Ranking Machine And İts Application To To Rogid Localization Of Deformable Objects, IEEE International Conference On Image Processing ICIP 2009, Cairo Egypt, Nov.7-10 2009,

39. E. Dule, M.Gökmen, M.S. Beratoglu, A Convenient Feature Vector Construction For Vehicle Color Recognition, Proc. Of The 11th WSEAS Int. Conf. Evolutionary Computing And 11th WSEAS İnternational Conference On Fuzzy Systems, Wisconsin, USA, 2010, Pp.978-960-474-195-3
40. B. Tunc, V.Daglı, M. Gökmen, Robust Face Recognition With Class Dependent Factor Analysis," Int. Conf. On Biometrics (IJCB), 2011, 11-13 Oct 2012, Pp 1-6
41. A. Akyol, M.Gökmen, "Face Super Resolution İn Reduced Spaces By Using Shape And Texture," MVA2011 IAPR Conference On Machine Vision Applications, June 13-15, 2011, Nara, JAPAN 14-6 438
42. A. Akyol, M.Gökmen," Efficient Face Hallucination By Using Shape And Texture Dependency. ICIP 2011: 1129-1132. 2012
43. E. Sarıyanidi, Volkan Dağlı, Salih Cihan Tek, Birkan Tunç, Muhittin Gökmen, "Local Zernike Moments: A New Representation For Face Recognition, Proc. Int. Conf. On Image Processing, ICIP2012, Orlando, USA
44. E. Sarıyanidi, B. Tunç, M. Gökmen, LZM in action: realtime face recognition system, Computer Vision–ECCV 2012. Workshops and Demonstrations, 2012
45. SC Tek, M Gökmen, GPU Accelerated Real-time Object Detection on High Resolution Videos using Modified Census Transform. Int. Conf. On Computer Vision Theory and Application, VISAPP (1), 685-688, 2012.
46. E Sariyanidi, H Gunes, M Gökmen, A Cavallaro, Local Zernike moment representations for facial affect recognition Proc. of British Machine Vision Conf, 2013.
47. G Özbulak, M Gökmen, A rotation invariant local Zernike moment based interest point detector, Seventh International Conference on Machine Vision (ICMV) 2014.
48. M Uzun-Per, M Gokmen, Face recognition with a novel image representation: Local Walsh-Hadamard Transform, Visual Information Processing (EUVIP), ≈ 5th European Workshop on, 1-6æ, 2014
49. Tolga Alasag, Muhittin Gokmen, Face Recognition in Low-resolution Images by Using Local Zernike Moments, Proceedings of the International Conference on Machine Vision and Machine Learning Prague, Czech Republic, August14-15, 2014
50. Özbulak G., Gökmen M. (2015). Robust Interest Point Detection by Local Zernike Moments. 10th International Conference on Computer Vision Theory and Applications (VISAPP), March 11-14, Berlin, Germany.
51. B.S.A. Gazioğlu, M. Gökmen, Facial expression recognition from still images, Int. Conf. on Augmented Cognition, 413-428, 2017,

52. M.M. Kalayeh, E. Basaran, M. Gökmen, M.E. Kamasak, M. Shah, “Human semantic parsing for person re-identification,” Proc. of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2018, pp. 1062-1071.
53. M Gokmen, E Sariyanidi, L Yankowitz, CJ Zampella, RT Schultz, B Tunç, “Detecting Autism from Head Movements using Kinesics”, in Internatıonal Conference on Multimodal Interaction (ICMI ’24), November 04–08, 2024, San Jose, Costa Rica. ACM, New York, NY, USA, 5 pages. 2024. https://doi.org/10.1145/3678957.368571

Books:

M.Gökmen, Edge Detection Using Regularization Theory, UMI-Bell&Howell Information Company, Ann Arbor Michigan, 1990, (Order Number 9120468) (Basılmış Doktora Tezi)

National Conference Papers (in Turkish):

1. T.Acar ve M. Gökmen, "Uyarlanır Düzleme ve Yırtılabilir Zar Modelinin Ayrıt Saptama Açısından Karşılaştırılması," 1. Sinyal İşleme ve Uygulamaları Kurultayı,İstanbul, 21-22 Nisan 1993, pp. 111-116.
2. N.Apak ve M. Gökmen, "Sayısal Hücre Resimlerinin Nicel Analizi ve Kodlanması," 1. Sinyal İşleme ve Uygulamaları Kurultayı , Istanbul, 21-22 Nisan 1993, pp. 117-122.
3. T.Acar ve M. Gökmen, "Yırtılabilir Zar Modeliyle Resim Onarımı," 5. Elektrik Mühendisleri Odası Konferansı, KTÜ Trabzon, 13-18 Eylül, 1993, pp.247-252.
4. A.Gençata ve M. Gökmen, "Zar ve Levha Modellerinden oluşan Karma Model Yardımıyla Yüzey Kurma ve Ayrıt Saptama," 2. Sinyal Işleme ve Uygulamaları Kurultayı, Gökova Muğla, 8-9 Nisan, 1994, pp.1-6.
5. T.Acar ve M. Gökmen, "Ağırlıklı Toplama Yöntemiyle Çok Ölçekli Ayrıt Birleştirme," 2. Sinyal Işleme ve Uygulamaları Kurultayı, Gökova Muğla, 8-9 Nisan, 1994, pp. 65-70.
6. A.Gültekin ve M. Gökmen, "Görüntü Tonlarından Şekil Elde Etme," 2. Sinyal İşleme ve Uygulamaları Kurultayı, Gökova Muğla, 8-9 Nisan, 1994, pp. 71-76.
7. C. K. Sandalcı ve M. Gökmen, "Parmakizi Görüntülerinin Sınıflandırılması," 2. Sinyal Işleme ve Uygulamaları Kurultayı, Gökova Muğla, 8-9 Nisan, 1994, pp. 134-139.
8. M.Gökmen, "λ-τ Uzayında Görüntü Gösterilimi ve Genelleştirilmiş Ayrıt Saptayıcı," 3. Sinyal İşleme ve Uygulamaları Kurultayı, A:Görüntü İşleme , Kapadokya Nevşehir, 26-28 Nisan, 1995, pp. 19-24.
9. Ö.Doğan ve M. Gökmen, "Hızlı çokgen Yaklaşımı ile Yapay Sinir Ağı Temelli Nesne Tanıma," 3. Sinyal İşleme ve Uygulamalar Kurultayı , A: Görüntü İşleme, Kapadokya Nevşehir, 26-28 Nisan, 1995, pp. 99-104.
10. A.Gençata, T. Acar ve M. Gökmen, "Yırtılabilir Zar Modeliyle Görüntü Sıkıştırma," 3. Sinyal İşleme ve Uygulamalar Kurultayı, A: Görüntü İşleme, Kapadokya Nevsehir, 26-28 Nisan, 1995, pp. 117-122.
11. B.Çınarkaya, A.G.Erkman ve M. Gökmen, " Sözdizimsel Yöntemle Görüntü Tanıma," 3. Sinyal İşleme ve Uygulamaları Kurultayı, A: Görüntü İşleme, Kapadokya Nevşehir, 26-28 Nisan, 1995, pp. 193-197.
12. I.Atalay ve M. Gökmen, "Özyüz yöntemiyle yüz tanıma," 4. Sinyal Işleme ve Uygulamaları Kurultayı, Kemer Antalya, 1996, pp. 151-156.
13. B.Kurt ve M.Gökmen, "Kırkayak Modeline Dayalı Görüntü Sıkıştırma," 5. Sinyal Işleme ve Uygulamaları Kurultayı, Kuşadası İzmir, 1-3 Mayıs 1997, vol.1, pp. 307-312.

14. B.Kurt, T. Demiral ve M.Gökmen, "Ayrıta Dayalı Etkin Bir Görüntü Gösterimi," 6. Sinyal Işleme ve Uygulamaları Kurultayı, Kızılcahamam Ankara, 1998, vol.1, pp. 259-264.
15. B.Kurt ve M.Gökmen,"İki Boyutlu Genelleştirilmiş Ayrıt Saptayıcı," 7. Sinyal Işleme ve Uygulamaları Kurultayı, Hotel Bilkent Ankara, 16-19 Haziran 1999.
16. A.Yılmaz ve M.Gökmen,"Öztepeler yöntemi ile Yüz Tanıma," 7. Sinyal Işleme ve Uygulamaları Kurultayı, Hotel Bilkent Ankara, 16-19 Haziran 1999.
17. T.Demiral ve M.Gökmen, "Benek ve Geometri Benzerliği Kullanılarak Aşamalı Çevrit Eşleme," 7. Sinyal Işleme ve Uygulamaları Kurultayı, Hotel Bilkent Ankara, 16-19 Haziran 1999.
18. A.Yılmaz ve M.Gökmen,"Yüz Tanıma Algoritmalarına Yönelik Aydınlanma Analizi," 8. Sinyal Işleme ve Uygulamaları Kurultayı,Belek Antalya, 12-14 Haziran 2000, pp. 118-121.
19. B.Kurt ve M.Gökmen,"Sahne Değişiminin Geometri Tabanlı Olarak Saptanması," 8. Sinyal Işleme ve Uygulamaları Kurultayı,Belek Antalya, 12-14 Haziran 2000, pp. 512-517.
20. Z. Erdem, F.Kurugöllü, M.Gökmen, "Genelleştirilmiş Ayrıt Saptayıcının Ardışıl Olarak Gerçeklemesi", 8. Sinyal Işleme ve Uygulamaları Kurultayı,Belek Antalya, 12-14 Haziran 2000, pp. 45-50.
21. A. Çapar, Ö.Kılıç, K. Taşdemir, M. S. Beratoğlu, M.Gökmen "Elyazısı Tanıma Sistemlerinde Öznitelik Çıkarma Ve Sınıflandırma Yöntemlerinin Karşılaştırması" URSI TÜRKİYE 2002, Sayfa 210-213.
22. A. Çapar, M. S. Beratoğlu, K.Taşdemir, Ö.Kılıç, M.Gökmen, "İTÜ Araç Plaka Tanıma Sistemi", 11. Sinyal İşleme ve İletişim Uygulamaları Kurultayı (SİU) 2003 İstanbul, Sayfa 371-374.
23. M.S. Beratoğlu, M. Gökmen "ITU VIP – Görsel Görüntü İşleme Ortamı", 11. Sinyal İşleme ve İletişim Uygulamaları Kurultayı (SİU) İstanbul 2003, Sayfa. 371-374.
24. F. Kahraman, M. Gökmen, "Gabor Süzgeçler Kullanılarak Taşıt Plakalarının Yerinin Saptanması" 11. sinyal İşleme ve İletişim Uygulamaları Kurultayı, Istanbul 2003.
25. A. Çapar, A. Ayvacı, F.Kahraman, H.Demirel, M. Gökmen "El Yazısı Tanıma Sistemlerinde Doğrusal Boyut İndirgeme Yöntemleri" IEEE 12. Sinyal İşleme ve İletişim Uygulamaları Kurultayı (SİU) 2004, Kuşadası pp. 611-615
26. F. Kahraman, A.Çapar, A.Ayvacı, H.Demirel, M.Gökmen "Elyazısı Verileri Üzerinde YSA ve DVM' nin Sınıflandırma Başarımlarının Karşılaştırılması" IEEE 12. Sinyal İşleme ve İletişim Uygulamaları Kurultayı (SİU) 2004, Kuşadası pp. 615-619
27. A.Çapar, A.Ayvacı, F. Kahraman, H. Demirel, M. Gökmen, El Yazısı Tanıma Sistemlerinde Doğrusal Boyut İndirgeme Yöntemleri, , SİU 2004 Konferans Kitabı, Kuşadası.
28. H. Demirel, S. Şener, M. Gökmen Geliştirilmiş Yayılı İzge Yöntemi ile Video Damgalama, SIU 2004, Kuşadası.
29. Çapar, M. Gökmen, Öz-Uzay Dağılım Enerjili Markov Rasgele Alanlarıyla Nesne Yeri Saptama, IEEE 13. Sinyal İşleme ve İletişim Uygulamaları Kurultayı, 16-18 Mayıs 2005, Kayseri
30. F.Kahraman, M.Gökmen, Aktif Görünüm Model Temelli Yüz Tanıma, IEEE 13. Sinyal İşleme ve İletişim Uygulamaları Kurultayı, 16-18 Mayıs 2005, Kayseri
31. F.Kahraman, M.Gökmen Bakış Açısından Bağımsız Gürbüz Plaka Tanıma Sistemi, IEEE 13. Sinyal İşleme ve İletişim Uygulamaları Kurultayı, 16-18 Mayıs 2005, Kayseri
32. F.Kahraman, M.Gökmen, Işıktan Bağımsız ve Bileşen Temelli Yüz Hizalama Yaklaşımı, IEEE 14. Sinyal İşleme ve İletişim Uygulamaları Kurultayı (SİU'06), Belek, Antalya, 17-19 Nisan 2006

33. F.Kahraman, B. Kurt, M.Gökmen, Seçkin Ayrıtlar Kullanılarak Üç-Bant Modelleme ile Yüz Hizalama, IEEE 14. Sinyal İşleme ve İletişim Uygulamaları Kurultayı (SİU'06), Belek, Antalya, 17-19 Nisan 2006.

34. A.Çapar, M.Gökmen, Şekil Tabanlı Hızlı Yürüme Yöntemi ile Nesne Bölütleme ve Tanıma, IEEE 14. Sinyal İşleme ve İletişim Uygulamaları Kurultayı (SİU'06), Belek, Antalya, 17-19 Nisan 2006

35. Binnur Kurt, Abdulkerim Capar,Muhittin Gokmen, Affine Invariant Shape Descriptors; Signal Processing and Communications Applications, 2007. SIU 2007. IEEE 15th, 11-13 June 2007 Page(s):1 – 4
36. Erdem Guneş, Muhittin Gokmen, Gabor Filters Based Segmentation For Monaural Sound Segregation, Signal Processing and Communications Applications, 2007. SIU 2007. IEEE 15th, 11-13 June 2007 Page(s):1 – 4
37. Binnur Kurt, Muhittin Gokmen, Amaca Yönelik Ayrıt Saptama, Signal Processing and Communications Applications, 2008. SIU 2008. IEEE 16th, 20-22 Nisan 2008 Page(s):1 – 4
38. B. Tunç, M. Gökmen, Çok kipli altuzay Yaklaşımı ile Yüz Tanıma, Signal Processing and Communications Applications, 2010 SIU 2010 IEEE 18th 22-24 Nisan 2010. Pp.391-394
39. M. Kaplan, M. Gökmen, Yüksek Başarımlı Otomatik Trafik Denetim Sistemi, Signal Processing and Communications Applications, 2010 SIU 2010 IEEE 18th 22-24 Nisan 2010.
40. Evangelos Sarıyanidi, Salih Cihan Tek and Muhittin Gökmen, "Seyrek örnekleme Yöntemiyle Etkin Yüz Saptama", in Sinyal İşleme ve İletişim Uygulamaları Kurultayı (SIU) 2011, in Antalya, April 2011 http://www.cvip.itu.edu.tr/media/3057536.pdf
41. Birkan Tunç, Volkan Dağlı and Muhittin Gökmen, "Sınıfa özel Etmen çözümlemesiyle Nesne Tanıma", in Sinyal İşleme ve İletişim Uygulamaları Kurultayı (SIU) 2011, in Antalya, April 2011 http://www.cvip.itu.edu.tr/media/1776252.pdf
42. Mahmut Meral, M. Gökmen" lambda tau uzayında yüz tanıma," in Sinyal İşleme ve İletişim Uygulamaları Kurultayı (SIU) 2011, in Antalya, April 2011 http://www.cvip.itu.edu.tr/media/1776252.pdf
43. Evangelos Sarıyanidi, Volkan Dağlı, Salih Cihan Tek, Birkan Tunç, Muhittin Gökmen, "Yüz Tanıma için yeni bir görüntü gösterimi:Yerel Zernike Momentleri," Sinyal İşleme ve İletişim Uygulamaları Kurultayı (SIU) Fethiye Muğla, Nisan 2012.
44. E Basaran, M Gokmen, Traffic sign classification with Quantized Local Zernike Moments, Signal Processing and Communications Applications Conference (SIU), 2013 ,
45. BS Akkoca, M Gokmen, Facial expression recognition using Local Zernike Moments, Signal Processing and Communications Applications Conference (SIU), 2013
46. SE Kahraman, M Gokmen, Face pair matching with Local Zernike Moments and L2-Norm metric learning, Signal Processing and Communications Applications Conference (SIU), 2014.
47. E Basaran, M Gokmen, A novel face recognition method based on Local Zernike Moments, Signal Processing and Communications Applications Conference (SIU), 2014.
48. BS Coban, M Gokmen, Gender classification with Local Zernike Moments and local binary patterns, Signal Processing and Communications Applications Conference (SIU), 2014.
49. BS Akkoca, M Gokmen, Facial expression recognition from static images, Signal Processing and Communications Applications Conference (SIU), 2014
50. BS Akkoca, M Gokmen, Automatic smile recognition from face images, Signal Processing and Communications Applications Conference (SIU), 2015 ...
51. M Uzun-Per, M Gokmen, Face recognition with Local Walsh-Hadamard Transform around landmarks, Signal Processing and Communications Applications Conference (SIU), 2015 ...
52. G Ozbulak, M Gokmen, Corner detection by Local Zernike Moments, Signal Processing and Communications Applications Conference (SIU), 2015 ...

53. G Vural, M Gokmen, Facial feature detection using Conditional Regression Forests, Signal Processing and Communications Applications Conference (SIU), 2015 .
54. E Başaran, M Gökmen, Face recognition with Local Zernike Moments features around landmarks," 24th Signal Processing and Communication Application Conference (SIU-2016), 2016, pp.2089-2092.

Edited Conference Proceedings (in Turkish):

1. M.Gökmen, E. Panayırcı, D.Demir, Ö. Yetiş, Editor, Kurultay Kitapı, A: Görüntü İşleme, 3. Sinyal İşleme ve Uygulamaları Kurultayı, Kapadokya Nevsehir, 26-28 Nisan, 1995.

2. M.Gökmen, E. Panayırcı, D.Demir, Ö. Yetiş, Editor, Kurultay Kitapı, B:Işaret İşleme, 3. Sinyal İşleme ve Uygulamaları Kurultayı, Kapadokya Nevşehir, 26-28 Nisan, 1995.

Scientific Reports:

Technical Reports:

1. R.W. Hall and M. Gökmen, "Rectangular/Hexagonal Tessellation Transforms and Parallel Shrinking," Technical Report, Department of Electrical Eng., University of Pittsburgh, TR-SP-90-004, 1990.

2. M.Gökmen and C. C. Li, "Edge Detection Using Refined Regularization," Technical Report, Department of Electrical Eng.,University of Pittsburgh, TR-SP-91-02, 1991.

3. M.Gökmen and C. C. Li, "Multiscale Edge Detection Using First Order R-Filter". Technical Report, Department of Electrical Eng., University of Pittsburgh, TR-SP-91-04, 1991.

Project Reports:

4. DARPA: Wavelet Transforms and Parallel Image Processing, 1989-1991.

5. NSF : US-TURKEY Collaborative Research on Edge Detection and Surface Reconstruction Based on $\lambda\tau$-Space Representation, 1993-1995.

6. TÜBİTAK: (EEEAG-154) Solving Computer Vision problems such as Edge Detection, Surface Reconstruction, Image & Video Compression, Shape from Shading, fingerprint compression by means of Regularization Theory 1995-1997.

7. TÜBİTAK: EEEAG-104E121, "Multi-purpose robust face recognition ", 2007-2008

Thesis Supervision:

Ph.D. thesis completed:

1. Attila Gültekin, "Adaptive Shape from Shading," İTÜ Kontrol-Bilgisayar Müh.Böl., ITU FBE, June 1997.
2. İlhan Kubilay Yalçın, "3D Model Based Sthocastic tracking of License Plates in Video Sequences," ITU FBE , October 2007
3. Binnur Kurt, "Goal-Oriented Edge Detection", ITU FBE , Oct., 2007
4. Fatih Kahraman, Robust Face Alignment using Active Appearance Model (in Turkish), ITU BE, October 2009.
5. Abdülkerim Çapar, "Object Segmentation and Recognition Using Gradient based descriptors and shape driven fast marching methods, ITU FBE,June 2010,
6. Birkan Tunc, "Robust Face Recognition on Nonlinear Manifolds," ITU BE June 2012,
7. Aydın Akyol,"Efficient Techniques for the Single Frame Super-resolution Reconstruction of Intensity Images,", ITU FBE, June 2012.
8. Meryem Uzun Per, "Face recognition with Local Walsh Transform,", ITU FBE, February 2018
9. Emrah Başaran, "Face Recognition and person identification for person recognition," İTÜ FBE February 2020.

M.S. Thesis completed:

1. Neşe Apak, Sayısal Hücre Görüntülerinin Kodlanması ve Nicel Analizi, 1993.
2. Tolga Acar, Solving Image Processing Problems by Using Nonstandart Regularization, 1994.
3. Ayşegül Gencata, Yırtılabilir Zar Modeli Yardımıyla Görüntü Sıkıştırma, 1995.
4. Ilker Atalay, Face Recognition Using Eigenfaces, 1996.
5. Ilker Ersoy, Model Based Compression of Fingerprint Images, 1996.
6. Binnur Kurt, Image Compression Based on Centiped Model, 1997.
7. Hüseyin Ergün, Dalgacık Dönüşümündeki Yerel Tepe Değerlerinden Görüntü Geri Çatma ve Sıkıştırma, 1997.
8. Ercan Özyıldız, Face Recognition, 1997.
9. Sinan Konya, Texture Features for Browsing and Retrieving of Image Data, 1998.
10. Bahar Colakoğlu, Yılan:Aktif Çevrit Modeli, 1998.
11. Barboros Günay, Türk Halılarının Görüntü Veritabanı Kullanılarak Saklanması ve Sorgulanması, 1998.
12. Özlem İnandık, Öznitelik Tabanlı Otomatik Parmakizi Eşleme, 1998.
13. Pınar Kayacı, Video Segmentation, 2000.
14. Oğuz İçoğlu, Indexing & Browsing of Multimedia Content Data with MPEG-7, Haziran 2001.
15. Muhammet Sebul Beratoğlu, Araç Plaka Yerinin Saptanması, 2003.
16. Kadim Taşdemir, Feature and Decision Fusion for Character Recognition, December 2003.
17. Hakan Demirel, Digital Image and Video Watermarking, July, 2004

18. Rupen Melkisetoğlu, Quadtree Based Rendering and Paging Algorithms for Very Large Scaled Terrains, December 2004,
19. Özcan Çavuş, Face Recognition with Kernel Fisher Discriminant Analysis, December 2005
20. Kürşat Ağır, Null Space Approach of Fisher Discriminant Analysis for Face Recognition, December 2005
21. Tuba Şirin, Çoklu Spektral Uzaktan Algılama Görüntülerinde Bölütleme Temelli Karma Ayrıt Saptama Algoritması, Haziran 2005
22. Alper Ayvacı, Multi-scale Vessel Boundary Detection, May 2006
23. Erald Vuçini, Face Recognition Under Varying Illumination, May 2006.
24. Özcan Çavuş, Face Recognition with Kernel Fisher Discriminat Analysis, 2006
25. Emre Bertan, Automatic 3D Roof Reconstruction using Digital Cadastral Map, Architectural Knowledge and an Aerial Image, June 2006.
26. Mahmut Meral, Embedded Speech Recognition, January, 2006
27. Kadir Kırtaç, Face Recognition using Gabor Features and Nearest-Neighbor Discriminant Analysis (NNDA) , June 2008
28. Barış David Ekinci, Ball Tracking in Tennis Videos, May, 2008
29. Mehmet Şerif Bayhan, Scale and Pose Invariant Real-Time Face Detection and Tracking, May 2008
30. Özgür Bulkan, Embedded DSP Based License Plate Localization, June 2008
31. Merve Can Kuş, Colored Traffic Sign Recognition Using Scale-Invariant Feature Transform and Color Classification Method, June 2008
32. Kıvılcım Helhel, Face Recognition Using Spherical Harmonics, May 2009,
33. Ceyda Karaboyacı, Geometrik Özellik Temelli Otomatik Yüz İfadesi Analizi, Haziran 2009
34. Hale Dodurgalı, Karınca Kolonisi Optimzasyonu ile Eğitilmiş Çok Katmanlı Yapay Sinir Ağı ile Sınıflandırma, Aralık 2009
35. Erida Dule, Vehicle Color Recognition, December 2009.
36. Mehmet Kaplan, An Automated Robust Vehicle Detection and Tracking System for Low Resolution Traffic Video Sequences, 2010,
37. Volkan Dağlı, Face Recognition using 3D Reconstruction and Albedo Estimation, 2011.
38. Salih Cihan Tek, Fast Face Detection And Recognition On Graphics Processing Unit (GPU), 2011.
39. Nil Seri, Near-Infrared Image Based Face Recognition, 2012
40. Merve Astekin, Code Cloning in simulation softwares, 2012.
41. Emrah Başaran, Traffic Signs Classification using Quantised Local Zernike Moments, 2013
42. Tolga Alasağ, Classification of the low-resolution face images by local zernike moments method, 2014
43. Bilge Süheyla Akkoca, Facial Expression recognition from Still Images, 2014
44. Şeref Emre Kahraman, Face Matching using Local Zernike Moments and Metric Learning Methods, 2015
45. Gökhan Özbulak, R-LZMF: Robust Local Zernike Moment Based Features, 2015
46. Gencer Vural, Detecting Face Landmarks using Conditional Regression Forests, 2015

Courses :

ITU:

- Image Processing, Computer Engineering, Graduate Program,
- Computer Vision, Computer Engineering, Graduate Program
- Advanced Topics in Computer Vision, Computer Science, Graduate Program

- Guided Research, Computer Science, Graduate Program
- Artificial Intelligence, Computer Engineering B.S. Program
- Software Engineering, Computer Engineering B.S. Program
- Computer Programming, Computer Engineering B.S. Program
- Computer Projects, Computer Engineering B.S. Program
- Introduction to Sci. and Eng. Comp. Prog., Computer Engineering B.S. Program

MEF University:

- COMP100-Introduction to Computer Engineering, Computer Engineering B.S. Program
- COMP204-Programming Studio, Computer Engineering B.S. Program
- COMP303-Analysis of Algorithms, Computer Engineering B.S. Program
- COMP482-Computer Vision, Computer Engineering B.S. Program
- ITC 532-Computer Vision, Information Technologies M.Sc. Program

Organized Conferences:

- International Modeling and Simulation Conference, University of Pittsburgh PA, 1989
- The Tenth International Symposium on Computer and Information Sciences (ISCIS X) Kuşadası Turkey 1995.
- Signal Processing and Applications (SIU) Workshop, SİU'95, Kapodokya Türkiye, 1995.
- The 23rd International Symposium on Computer and Information Sciences (ISCIS 2008) Istanbul Turkey, General Chair.

Conference Technical Committee membership:

- International Modeling and Simulation Conference, University of Pittsburgh PA, 1989
- International Symposium on Computer and Information Sciences (ISCIS X) Kuşadası Turkey 1995.
- International Symposium on Computer and Information Sciences (ISCIS XI) Antalya 1996.
- International Symposium on Computer and Information Sciences (ISCIS XII) Antalya 1997.
- Signal Processing and Applications Workshop, SİU'94, Gökova Türkiye, 1994
- Signal Processing and Applications Workshop, SİU'95, Kapadokya Türkiye, 1995
- Signal Processing and Applications Workshop, SİU'96, Kemer, Antalya, 1996
- Signal Processing and Applications Workshop, SİU'97, Kuşadası, Antalya 1997

- Signal Processing and Applications Workshop, SİU'99, Kızılcahamam, 1999

Research Projects:

Completed Projects

- DARPA: Wavelet Transforms and Parallel Image Processing, 1989-1991
- NSF: US-TURKEY Collaborative Research on Edge Detection and Surface Reconstruction Based on $\lambda\tau$-Space Representation, 1993-1995.
- UNDP: (TUR/95/002) TURKEY-Indigenous Software Development Project 1997-1999
- TÜBİTAK(EEEAG-154) Solving Computer Vision problems such as Edge Detection, Surface Reconstruction, Image & Video Compression, Shape from Shading, fingerprint compression by means of Regularization Theory , 1995-1997.
- DPT: ITU Multimedia R&D Project, 1997-1999.
- ITU Research Fund: Contour Based Video Coding, 1998-1999
- SSM: Wargames Graphical Interface Software (HOGAY), 2001-2003
- TUBİTAK-MAM: GÜMSİS License Plate Recognition System for Turkish Customs. 2001-2003
- TÜBİTAK: Multi-purpose Robust Face Recognition", 2005-2007
- TÜBİTAK- Pose and Illumination Invariant Face Recognition with Manifold Learning, 2009-2012
- DPT: Development of Generic Technologies in IT (BÜTEG), 2004 -2005
- DPT: Advanced Technologies, Graduate Program Coordinator of Computer Science Program , 2001-2009
- TÜBİTAK TEYDEB- Integrated Vehicle Recognition, 2008-2009.
- İBB: İstanbul Metropolitan Municipality: Traffic Analysis from Low Resolution Cameras(240x320), Projem İstanbul, 2008-2009.
- TÜBİTAK: Pose and Illumination Invariant Face Recognition with Manifold Learning, 2009-2012
- TÜBİTAK-TEYDEB: Vehicle recognition and Video Analysis. (TATVAN), 2011-2012.
- İTÜ BAP: Real Time Face Detection and Recognition using GPU (in Turkish), 2010-2012
- İTÜ BAP: Face Recognition using Nonlinear Manifold Learning,(in Turkish) 2011-2013
- TÜBİTAK-TEYDEB: Multipurpose Face Recognition, (in Turkish), 2013-2014.
- TÜBİTAK 1001: Face and Object recognition using local transformations, 2013-2015.
- ITUNOVA-TTO: Camera Based localization in Covered Areas (in Turkish), 2015-2016

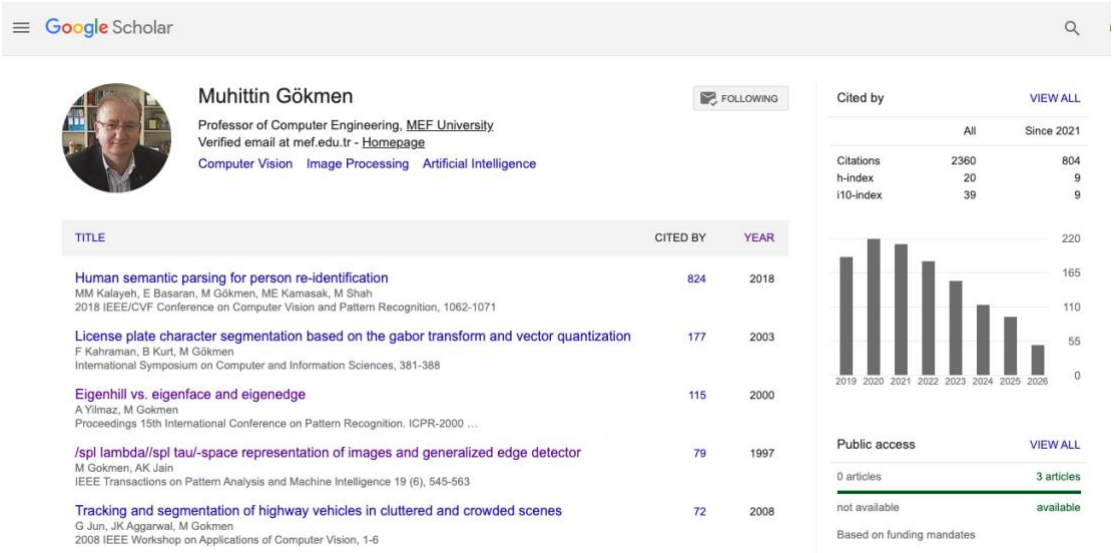
Seminars:

- The Robotics Institute, Carnegie Mellon University, Pittsburgh PA, ABD, 1990.
- Istanbul Technical University, Faculty of Electrical and Electronics, Istanbul, TURKEY, 1991
- TUBİTAK Marmara Research Center , Gebze TÜRKİYE, 1992
- Boğaziçi University, Department of Electrical Engineering, İstanbul TURKEY, 1992
- Bulgarian Academy of Science, Sophia, BULGARY, 1993
- Michigan State University, Department of Computer Science, Lansing MI, ABD 1994.
- Tutorial, IEEE International Conference on Image Processing, 2021, Anchorage, Alaska, USA

Journals served as Editor or Referee

- Machine Vision and Applications, International Journal, Springer, Associate Editor,
- IEEE Transaction on Pattern Analysis and Machine Intelligence, Reviewer
- IEEE Transaction on Circuits and Systems, Reviewer
- IEEE Transaction on Image Processing, Reviewer
- IEEE Transaction on Multimedia
- Journal of Mathematical Imaging and Vision, Reviewer
- TÜBİTAK ELEKTRİK - Journal of Electrical Engineering and Computer Sciences, Reviewer

Citations in International Journals



Citations: 2360
h-index: 20
i10-index:39